OPEN ITEMS CHECKLIST — part 4 (15 October 2026)

Changes since part 3 (11 October 2026). Every item was verified by assembling all four builds (`SERIALPOLL` 0/1 by `UNITTESTS` 0/1) and by the assembler unit tests and ANS tests run by the user in MAME.

Static temporaries in page zero reduced (stages 5a to 5e)

- ☑ **Working values moved from page zero onto the stacks.** Parameters stay on the data stack and are reached `U` -relative; locals live in short `LEAU` or `S` frames (they survive `THROW` , because `CATCH` / `THROW` restore `S`); `X` and `Y` are used as temporaries. Covers `CMOVE` , `FILL` , `COMPARE` , `SEARCH` , `TYPE` , `SPACES` , `ROLL` , `2>R` , `2R>` , `2R@` , `.S` , `WORDS` , `.R` , `U.R` , `D.R` (shared helper `RTYPE`), `DUMP` , `UNESCAPE` , `SUBSTITUTE` / `SUBCOPY` and `ACCEPT` , among others. `GLOBALS` now uses 127 of 256 bytes (was 255); `MVSCRATCH` is gone and `$0100-$0105` is unclaimed.
- ☑ **Unit-test harness defect (stage 5c).** The assembler unit tests run before `COLD` , with `CODEHERE` = 0, so `PAD` fell at `$80` in page zero and overwrote `TYPE` 's variables once they had been renumbered (found by the user with `SERIALPOLL=0` , `TSTDOTS`). `TSTRUNNER` now sets `CODEHERE` to `APPCODE` . No change to the Forth source.
- ☐ **Remaining page-zero temporaries** (not yet moved): the multiply and divide scratch (`MSCR` - `MSCR4` , `PRODHI` / `PRODLO` , `DIV*` , `UDHI` / `UDLO`) and a few others.
- ☐ **Idea, not started:** use the `DP` register as a byte-save location in inner loops where a byte is currently saved and restored.

String words (stages 5f to 5j)

Found by the optional String word tests (`27_optional_strings`), which had the right expectations.

- ☑ **`SEARCH`** returned the needle's length as `u3` . It now returns the characters remaining from the match, and an empty needle matches at the start. `TSTSEARCH1` had the same wrong expectation (3) and now expects 5.
- ☑ **`REPLACES`** held one registration. It is now a table in `SUBPOOL`

(64 bytes, `$0200-$023F`): a linked list of entries (link 2, name length 1, text length 1, capacity 1, name, text), headed by `SUBHEAD`, next free byte `SUBFREE`. It keeps its own copy of both strings. Redefinition is in place when the new text fits, otherwise a new entry (old space not reclaimed). Names are case sensitive; entries are not scoped by `MARKER`. Name or text over 255 throws -18, a full pool -8. Three 8-character pairs fit (21 bytes each); the ANS tests need 25 bytes. `APPVARS` moved to `$0240`.

- ☑ **`SUBSTITUTE`** no longer throws on error: it returns a negative `n` (-18 output overflow, -9 source and destination are the same buffer).
- ☑ **`BLANK`** added (`32 FILL`).
- ☑ **`CLR-REPLACES`** added (non-standard): empties the `REPLACES` table.
- ☑ **`S"`, `."` and `ABORT"` with an empty string** ran on past the closing quote, because `WORD` skips leading delimiters. They now parse with `PARSE` (helper `PARSEQUOTE`); over 255 characters throws -18. The 3-byte `CODEHERE` reserve they used to make is gone, so `WORDMAXCHARS` could be 3 larger; it was left alone (comment fixed in stage 5j).
- ☑ **New tests.** Assembler: `TSTBLANK`, `TSTCMOVEOVL`, `TSTCMOVEGTOVL`, `TSTCMOVELONG`, `TSTSEARCH3/4`, `TSTREPLSUBS3/4`, `TSTCLRREPL`, `TSTSQUOTE3`. ANS file: `BLANK`, `CMOVE`, `CMOVE>`, `SEARCH` u3, `SUBSTITUTE` errors, `CLR-REPLACES`.
- ☑ **Shadow files.** Added `ACCEPTW.5`, `UNESCAPEW.5`, `SUBSTITUTEW.5` (stage 5e references that had no entry), and put the stage 5f-5j entries in assembler order.

Memory map (stage 5h)

`BASEDICT` `$D498` - `$DCC3` (2092 bytes); `BASECODE` `$DCC4` - `$FD14` polled (8273 bytes) or `$FE53` interrupt-driven (8592); `SUBPOOL` `$0200` - `$023F`; `APPVARS` `$0240` - `$1FFF` (7616). ROM content 10453 bytes polled / 10772 interrupt-driven (unused 5675 / 5356 of the 16K part after the 256-byte `INOUT` shadow).

Documentation

- ☑ `ClaudeForth.docx` and preview PDF dated 15 October 2026: memory map and ROM sizes, glossary (`BLANK`, `CLR-REPLACES`; `SEARCH`, `REPLACES`, `SUBSTITUTE`, `S"`, `."`, `ABORT"`, `CMOVE`, `CMOVE>`, `WORD` revised), Section 5 items, regenerated source appendix (stage 5j), rebuilt index and page numbers. Memory-map SVG and PDF regenerated.

- □ **Timeline** on the title page not updated (hours since 11 October).